LECTURE 13

PROCURING GOODS AND SERVICES FOR IT

Knowledge Area	Process				
	Initiating	**Planning**	**Executing**	**Monitoring & Control**	**Closing**
Procurement Management		Plan Procurement Management	Conduct Procurements	Control Procurements	Close Procurements

What is Procurement?

- Sometimes you need to hire an outside company to do some of your project work. That is called **procurement**, and the outside company is called the Supplier/Vendor/Seller/ Contractor.
- **Procurement** is the act of obtaining goods or services, typically for business purposes.

PROCUREMENT MANAGEMENT PROCESSES

- Plan Procurement Management
- Conduct Procurements
- Control Procurements
- Close Procurements

TYPES OF PROCUREMENT CONTACT

Any contract between two or more parties to deliver services or goods is a procurement contract. You can divide procurement contracts into four categories:

1. Fixed-Price Contract
2. Cost Reimbursable Contract
3. Time and Materials Contract
4. Purchase Order Contract

As a project manager, it is your job to select a contract that satisfies your requirements.

Fixed-Price Contract

Many experts call a Fixed-Price contract a lump-sum contract. You use this contract when the scope of work is fixed. Once the contract is signed, the seller is contractually bound to complete the task within the agreed price and time. Therefore, the seller bears most of the risk. There is no price renegotiation unless the scope of work changes.

- Generally, outsourcing and turnkey procurement contracts are Fixed-Price. This contract is useful with a well-defined scope.

- Since the cost cannot change, this contract is suitable for controlling costs.

- Here, change in scope is a costly affair. Contractors often get the contract by bidding the lowest price and then try to generate extra revenue at any opportunity, such as an added scope.

- We have seen contractors quarrel with project managers regarding the scope. They agreed initially but later argue on small issues to raise the change request.

- Make sure the scope is well-defined and as detailed as possible; observe any changes in scope carefully.

You can divide a Fixed-Price contracts into three categories:

1. Firm Fixed Price (FFP)
2. Fixed Price Incentive Fee (FPIF)
3. Fixed Price with Economic Price Adjustment (FP-EPA)

Firm Fixed-Price Contract (FFP)

- This is the simplest type of procurement contract. The seller has to complete the job within an agreed amount of money and time. The seller is responsible for any increase in cost and they are legally bound to complete the task within the agreement.
- A Firm-Fixed-Price contract is mostly used in government or semi-government contracts where the scope of work is defined with high detail.
- This contract is easy to float and receive bids, and it is evaluated on a cost basis, which is fairly quick. Since the seller bears the risk, the cost is higher. If the scope is not clear, the seller and buyer may have disputes. Any deviation from the original scope can cost you a lot.

Example: The seller has to paint the whole building for 50,000 USD within 18 months.

Fixed-Price Incentive Fee Contract (FPIF)

- Although the price is fixed, the seller may receive an incentive if they perform well. This incentive lowers the seller's risk.
- The incentive can be tied to any project metrics such as cost, time, or technical performance.

Example: The contractor will receive an incentive of 10,000 USD if they achieve the first milestone on time.

Fixed-Price with Economic Price Adjustment Contracts (FP-EPA)

You use a Fixed-Price with Economic Price Adjustment Contract when the agreement is multi-year. This contract has a special provision that protects the seller from inflation.

Example: About 3% of the cost of the project will increase after a certain time based on the Consumer Price Index.

Largely used when

- There is high level of clarity on scope of project
- Fair amount of clarity on the type of work and the process required to achieve the project results
- Limited variability in project or high level of predictability
- Heavy focus on accomplishing something within a predetermined cost
- Buyer wants to control the cost overrun

Examples of Fixed Price:

- **Purchasing an Apartment**: When you buy an apartment, the seller often sets a fixed price. Once you agree on the price and sign the contract, you are obliged to pay that amount, regardless of any other factors or costs incurred by the seller.
- **Holiday Packages**: Travel agencies frequently offer holiday packages at a fixed price. This price typically includes flights, hotels, and other services such as guided tours or meals. No matter what the actual costs are for the agency, you pay the fixed price agreed upon when you book the package.
- **Off-the-Shelf Software Packages:** When buying pre-packaged software, you pay a fixed price for the product. This price doesn't change based on how much it cost the developer to make the software or how many hours they spent creating it.
- **Construction of a Road:** In this case, a government or organization might contract a construction company to build a road for a fixed price. The construction company estimates the total cost and agrees to complete the project for that price, regardless of the actual costs incurred during the construction process.

Cost Reimbursable Contract

- This is a cost-reimbursable contract where the buyer reimburses the seller for the seller's costs plus a fixed profit (fee). For example, the buyer reimburses the seller for work and materials costs. They agreed on a fixed fee of 10,000 USD, so this is pure profit for the seller.

- You use a Cost Reimbursable contract when there is uncertainty in the scope, or if the risk is higher. In this contract, the risk is on the buyer as they pay for all costs.

- This contract provides you with better cost control when you do not have a well-defined scope.

- Scope Creep is an inherent drawback of a Cost Reimbursement Contract when the requirements are unclear. The seller may try to increase the cost to increase the fee or reimbursement.

- You can minimize this through proper monitoring or capping the profit. For example, the maximum profit is 10% of the total cost.

You can divide Cost Reimbursable contracts into three categories:

1. Cost Plus Fixed Fee (CPFF)
2. Cost Plus Incentive Fee (CPIF)
3. Cost Plus Award Fee (CPAF)

<u>Cost Plus Fixed Fee Contract (CPFF)</u>

- The seller is paid for all incurred costs plus a fixed fee, regardless of their performance. The buyer bears the risk.

- Organizations use this contract with high-risk projects where bidders are not interested in competing.

- CPFF contracts keep the seller safe from risks.

Example: Total cost plus 25,000 USD as a fee.

<u>Cost Plus Incentive Fee Contract (CPIF)</u>

- In a Cost Plus Incentive Fee contract, the seller will be reimbursed for all costs plus an incentive fee based upon achieving certain performance objectives mentioned in the contract. This incentive will be calculated using an agreed-upon formula.

- Here, the risk lies with the buyer; however, it is lower than in the Cost Plus Fixed Fee.

- In a CPIF contract, the incentive is a motivating factor for the seller.

- Generally, an incentive is a percentage of the savings that buyer and seller share.

Example: If the project is completed under budget, the seller will receive 25% of the savings.

<u>Cost Plus Award Fee (CPAF)</u>

- Here, the seller is paid for their costs plus an award fee. This extra will be based on achieving satisfaction according to specified performance objectives described in the contract.

- Note: There is a difference between an incentive fee and award fee. An incentive fee is calculated based on a formula defined in the contract and is an objective evaluation. An award fee depends on the client's satisfaction and is evaluated subjectively. An award fee is not subject to an appeal.

Example: If the seller completes the task by meeting or exceeding quality standards based on their performance, the buyer may give an award up to 10,000 USD.

Largely used when:

- Limited clarity on scope
- High variability in projects
- Projects are abstract and are expected to evolve over the course of time
- Required specialized skills by the vendor/supplier.
- Focused on creating the value & experience for the customer rather than completing the project on budget

Examples of Cost Reimbursable

- **Event Decor:** This could be for creative or innovative projects where the exact costs can't be predicted in advance. For example, when hiring an event organizer for a large gathering, the organizer might need to make various purchases such as decorations, lighting, audio equipment, and more. As the client, you would reimburse the organizer for these costs.

- **Media Campaigns:** Media campaigns often involve various unpredictable costs. For instance, creating a marketing campaign might require hiring a graphic designer, buying ad space, or paying for video production. These costs can vary greatly depending on the specific needs of the campaign and are typically reimbursed by the client.

- **Research and Development Projects:** In R&D projects, it's often hard to estimate the total costs upfront due to the unpredictable nature of the work. Therefore, organizations often use cost-reimbursable contracts for these projects.

Time and Materials Contract:

A Time and Materials (T&M) contract is a type of contract that is a hybrid of both fixed-price and cost-reimbursable contracts. In a T&M contract, the client agrees to pay the contractor based upon the time spent by the contractor's employees and for materials purchased to complete the project, plus the contractor's profit.

Here are the main components of a T&M contract:

- **Time:** This is often billed on an hourly rate. The contractor tracks the time spent on the project and invoices the client accordingly.
- **Materials**: These are the physical inputs needed for the project, which can include everything from office supplies to heavy machinery. The client reimburses the contractor for the cost of these materials.

Largely used when:

- Moderate level of certainty in the project
- Reasonable visibility towards effort and type of resources required
- Limited predictability of amount of resources required
- Moderate variability as the process, material, people required are predictable

Examples of Time and Material

- **Customized Software Development:** When a company needs a software solution tailored to its specific needs, it might hire a developer or a team on a Time and Material basis. The final cost of the project may depend on several uncertain factors, such as the complexity of the task or any changes in project requirements, making a Time and Material contract more suitable.
- **Staff Augmentation:** If a business needs to expand its team temporarily for a project, they might hire additional staff for a specific time period on a Time and Material contract basis. The company would pay for the time these temporary employees work, plus any related costs, like recruiting or training expenses.
- **Consulting Services:** Many consulting services use Time and Material contracts. Consultants often work with clients on complex projects where the time required for problem-solving or strategy development can vary widely. The consultant charges for their time and any expenses incurred during the engagement.

Purchase Order (PO) Contract:

This contract is used to buy commodities.

Example: Buying 10,000 bolts of cloth at the cost of 1.00 USD

PROPOSAL EVALUATION TECHNIQUES

You are going to have to work closely with the seller to figure out if his proposal really is appropriate for the work. You need to be very careful before you choose someone to do the work.

- **RFI: Request for information** is when a project owner has a problem they need fixed but do not know the best way to address it, they may issue a request for information (RFI) to gather possible remedies from contractors.
- **RFP: Request for proposal** is when once the project owner has an idea of the project they would like to contract out, they can send out a request for proposal (RFP) from several contractors.
- **RFQ: Request for quotation** is related, but much more price-driven than the other two proposal types. The owner usually has selected the solution they want to use and just wants to know how much it will cost. There are not a lot of questions about qualifications to answer, as it is more about ***pricing***, ***payment options***, and ***delivery timelines***.

CORE STEPS OF PROCUREMENT PROCESS:

Procurement processes vary greatly depending on each company's structure and needs, but generally include the following ten core steps:

1. Identify which goods and services the company needs. First, a business must identify its requirements for a specific item or a service. This may be a new item that the company hasn't previously purchased, a restock of existing goods or a subscription renewal. This step typically involves delving into the nitty-gritty details of what the business needs, such as the precise technical specifications, materials, part numbers or service characteristics. At this stage, it's a good idea to consult all business departments affected by the purchasing decision to ensure the procured items accurately reflect the needs of each department.

2. Submit purchase request. When an employee or business group needs to procure a significant quantity of new supplies or services, they make a formal purchase request (also known as a purchase requisition). A purchase request notifies the company that a need exists, usually via department managers, purchasing staff or the financial team, as well as specifications such as price, time frame needed, quantity and other important things for the purchasing team to keep in mind. The department overseeing the purchase can then approve or deny the purchase request. If approved, the procurement team can proceed with selecting a vendor and making the purchase.

3. Assess and select vendors. With a clear list of requirements and an approved purchase request, now is the time to find the best vendor and submit a request for quote (RFQ) – this is what the purchasing team sends to potential suppliers in order to receive a quote – it is important to be as detailed as possible so you can compare apples to apples. Vendor assessment should focus not only on cost but also on reputation, speed, quality and reliability. Many companies consider ethics and social responsibility as well, since procurement is often intertwined with corporate identity. A retailer that prides itself on sustainability would stand to benefit from partnering with environmentally responsible suppliers, for instance.

4. Negotiate price and terms. A common best practice is to get at least three quotes from suppliers before making a decision. Examine each quote carefully and negotiate where possible. If you need to walk away from a deal, be sure that you have concrete alternative options. Once you've agreed on final terms, be sure to get them in writing.

5. Create a purchase order. Fill out a purchase order (PO) and send it to the supplier. The PO should be sufficiently detailed to identify the exact services or goods needed and to enable the supplier to fill the order.

6. Receive and inspect the delivered goods. Carefully examine deliveries for any errors or damage. Make sure everything is delivered as specified in the PO and that the quality meets or exceeds expectations.

7. Conduct three-way matching. Accounts payable should conduct three-way matching by comparing the purchase order, order receipt or packing list and invoice. The goal is to ensure the goods or services received match the purchase order and to prevent payment for unauthorized or inaccurate invoices. Highlight any discrepancies between the three documents and resolve issues before arranging payment.

8. Approve the invoice and arrange payment. If the three-way match is accurate, approve and pay the invoice. Businesses should strive to have a consistent invoice payment process through accounts payable that checks that payments match the invoice amount and due date. A standardized process can help make sure invoices are always paid on time, which can prevent late fees and build good relationships with suppliers.

9. Recordkeeping. It's important to maintain records for the entire procurement process, from purchase requests to price negotiations, invoices, receipts and everything in between. These records may be useful for multiple reasons. They help the company reorder goods at the right price in the future, as well as assist with auditing processes and calculating taxes. Clear, accurate records can also help resolve any potential disputes.

10. Close Procurement. When the work is done, you will close your contract out. You will make sure that the product that is produced meets the criteria for the contract, and that the contractor gets paid.

S. No.	Name of Process	Step(s) Include
1	Plan Procurement Management	1
2	Conduct Procurements	2,3,4 and 5
3	Control Procurements	6,7,8 and 9
4	Closed Procurements	10